

“We must **trust to nothing but facts:** these are **presented** to us **by nature** and **cannot deceive.**”

Antoine-Laurent de Lavoisier, 1790

vigorously than ordinary air. Back in his laboratory, Lavoisier replicated Priestley's experiment. He heated mercury calx and measured how much “pure air” (or “oxygen,” as he later named it) was released as the calx broke down into mercury. He also heated the mercury and measured how much oxygen was taken up as it changed into mercury calx again. The amount of oxygen taken up and released matched.

When he heated other substances in sealed containers, Lavoisier found that the mass a metal gained when heated was exactly equal to the oxygen lost from the air. As a result of these experiments Lavoisier started to form his law of conservation of mass, which stated that matter could change in form but could not be created or destroyed.

Air, water, and elements

Lavoisier observed that a lit candle in a sealed jar would extinguish after a time and, from his various experiments on combustion, he knew that oxygen had allowed the candle to burn. However, he was also aware that some gas remained in the jar and deduced that air must contain several gases. He called the other gas “azote,” meaning “without life.” (It was later called “nitrogen.”) Lavoisier also

investigated respiration, which he thought converted oxygen into energy in the form of heat and an azotic gas (carbon dioxide) in the same way that a burning candle does. Lavoisier placed a guinea pig in a container that was surrounded by ice and measured the heat and carbon dioxide that the animal produced while 2 pounds of ice melted. He then compared these results with the amount of heat produced by burning carbon until the same amount of carbon dioxide was given off as the guinea pig had produced. In this way, Lavoisier showed that respiration (breathing) was also a form of combustion.

By comparing the weights of substances before and after chemical reactions, Lavoisier determined the law of conservation of mass: matter only changes forms; it cannot be created or destroyed.

